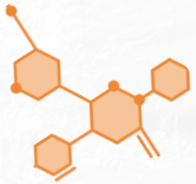
SUSTech Southern University of Science and Technology

# 基于过渡金属二硫族化物的光电传感器

Group 8 :周晓俊 章一民 王钰溪 秦子涵
仰先多 顾洋铭 水雨辰 梁程烨

2025/12/24





SUSTech Southern University of Science and Technology

# CATALOGUE


01 原理回顾

02 制造流程

03 分析优化

SUSTech Southern University of Science and Technology

# 01

# 原理回顾



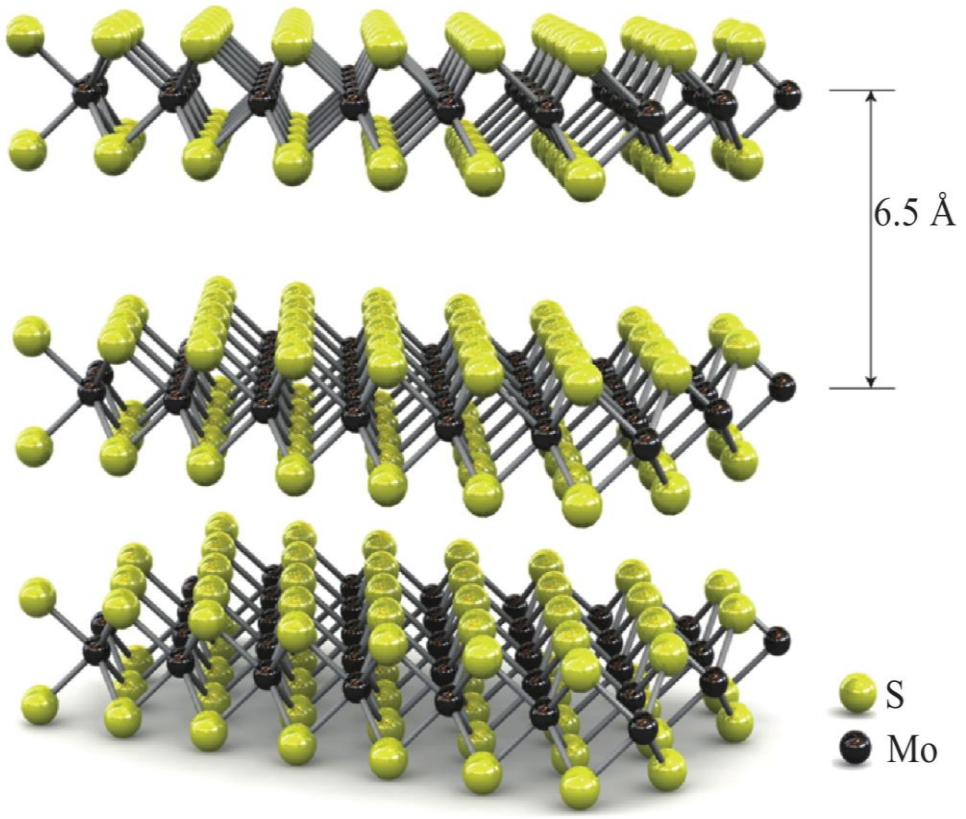
# 光电二极管

- 光电二极管核心部分是一个PN结
- 受到光照后，光生载流子在PN结内建电场和反向偏压的作用下分离，产生光电流，光的强度越大，反向电流也越大



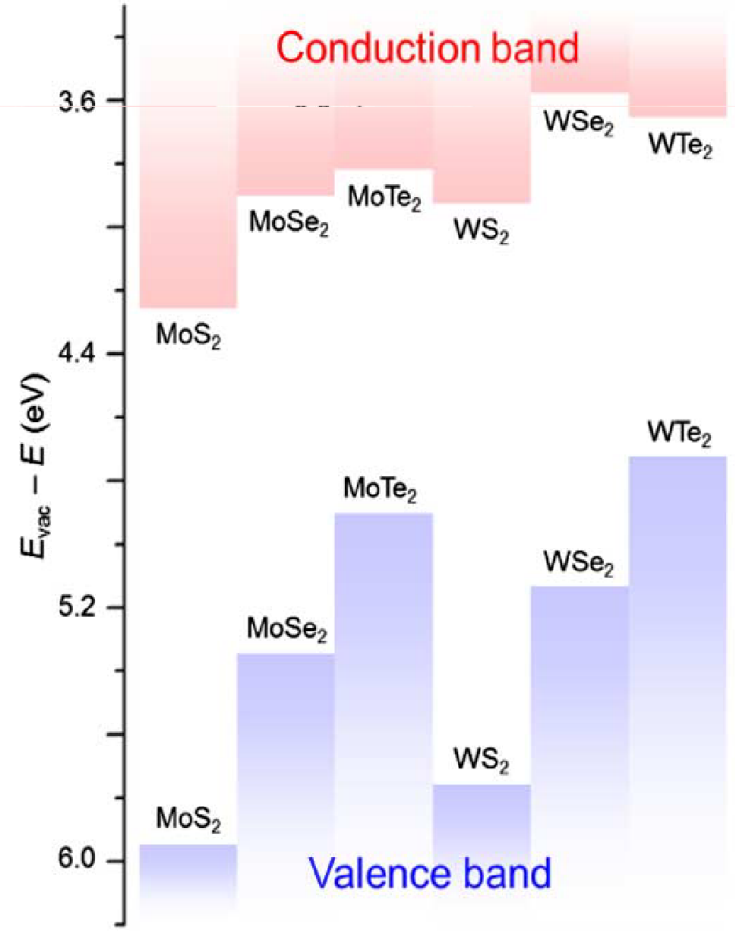
# 过渡金属二硫族化物

- 过渡金属二硫族化物（TMDCs）是一类具有层状范德华结构的二维半导体材料，组分为$MX_2$
- “三明治”层状结构：中间一层过渡金属原子（M），上下两层硫族原子（X）通过共价键紧密结合
- 单层厚度约 0.65-1 nm，层与层之间依靠弱范德华力连接，可轻松剥离为少层或单层



# 能带结构

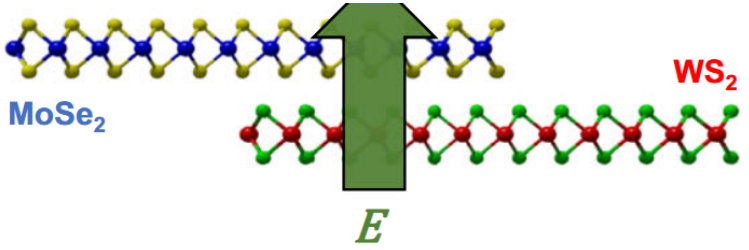
- 不同材料的导带底、价带顶位置差异显著，导致费米能级和带隙不同
- TMDCs材料的能带结构受到层数的影响。单层材料为直接带隙，多层则转变为间接带隙，带隙宽度随层数可调，层数越多，带隙越小

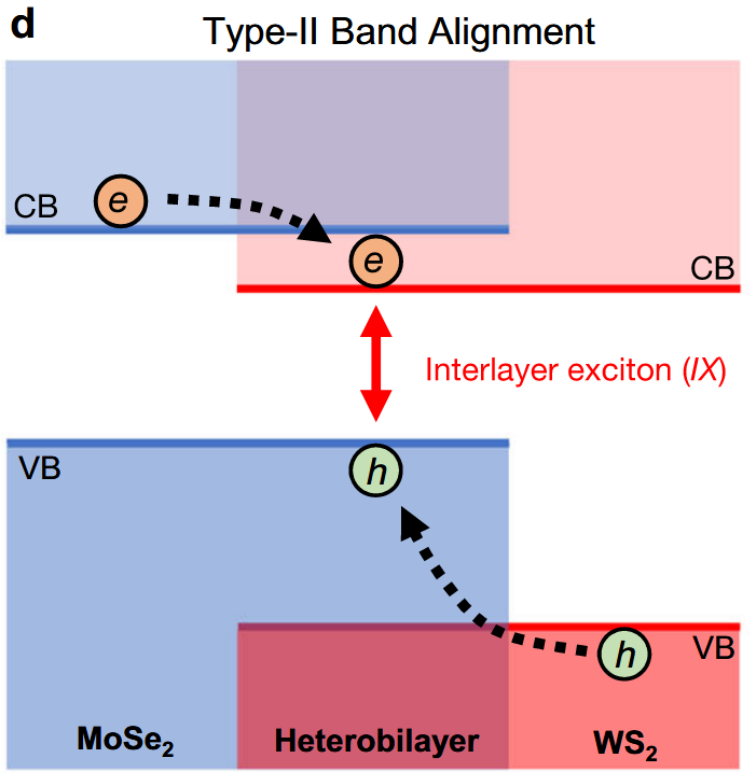


# TMDCs光电二极管

SUSTech Southern University of Science and Technology

- 将两种不同的TMDCs材料进行堆叠，由于其能带结构的不同，会形成二型能带（错开型）对齐，形成类PN结，在界面处形成内建电场，为电子和空穴分离提供动力



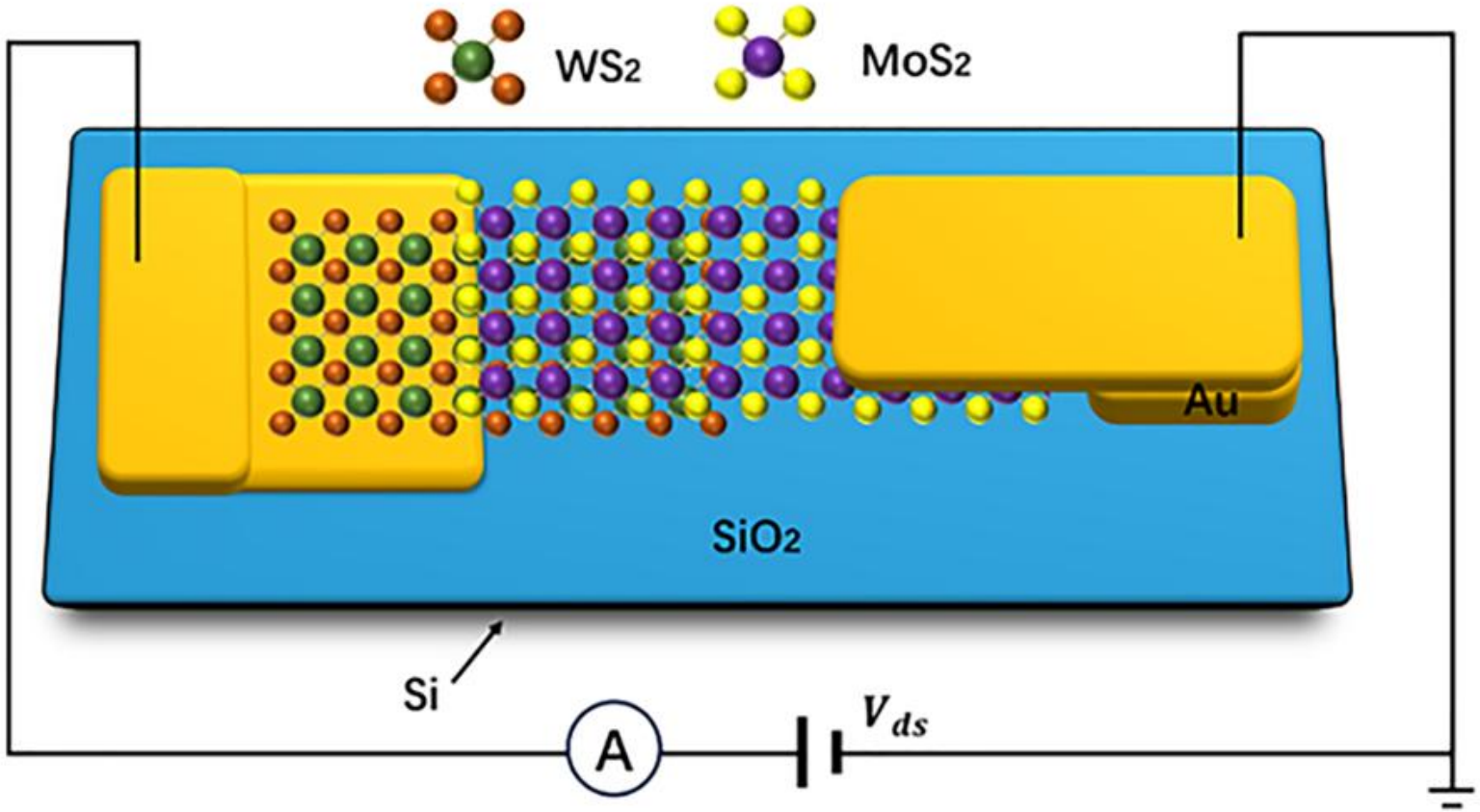
SUSTech Southern University of Science and Technology

# 02

# 制造流程



# 器件结构



# 衬底准备

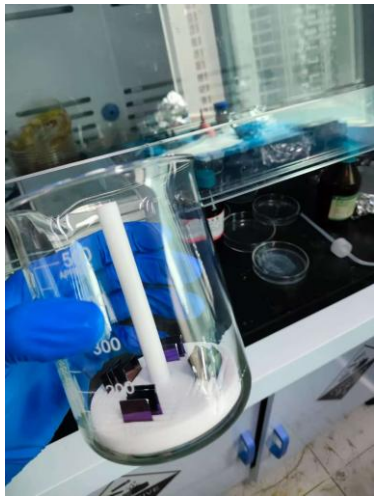
SUSTech Southern University of Science and Technology



4英寸晶圆



裂片






有机溶剂超声清洗

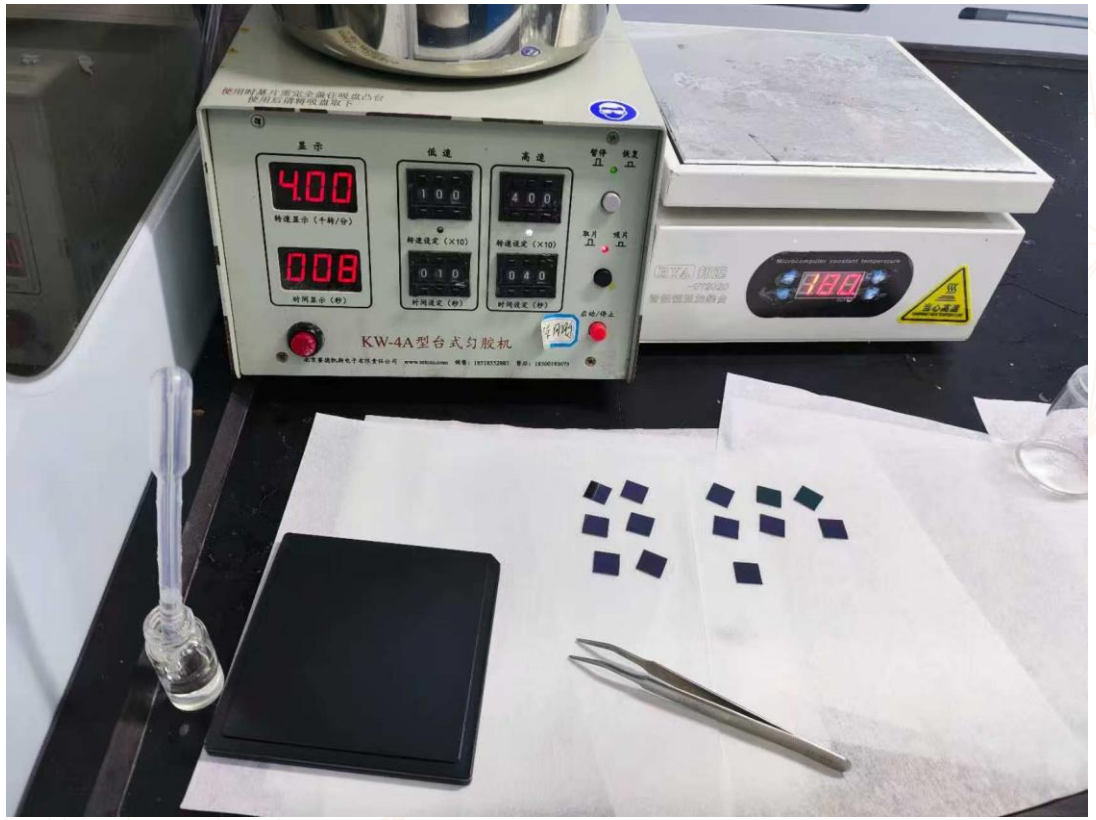



氩等离子体清洗

# 图案转移




旋涂PMMA电子束光刻胶

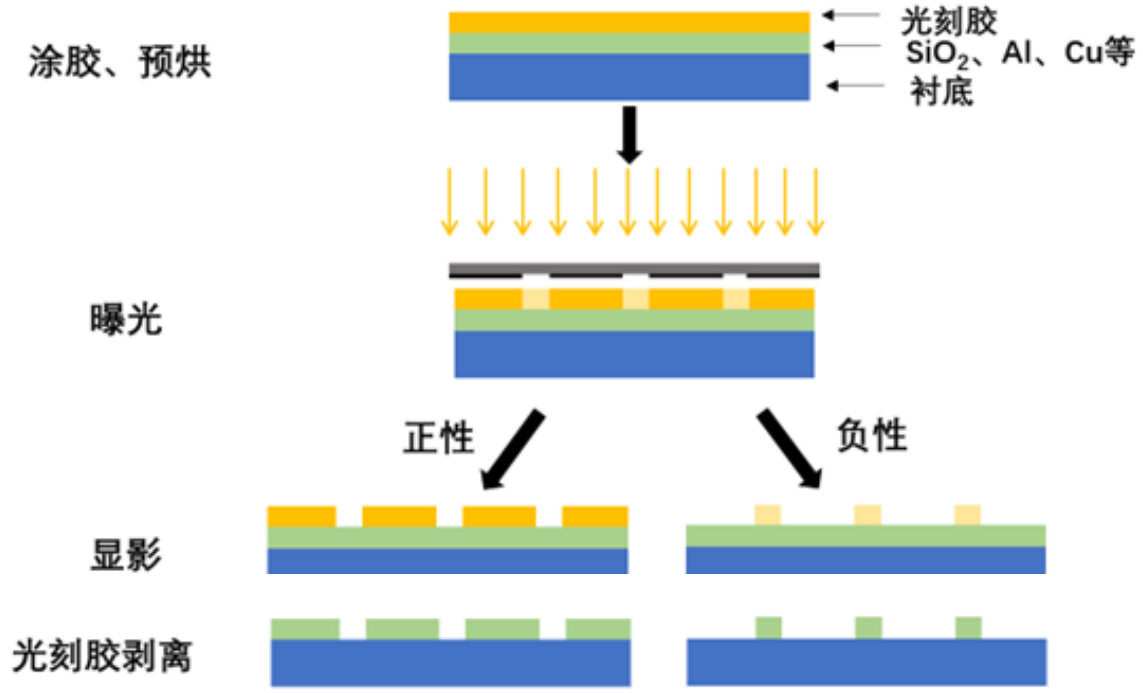



180°C烘胶

# 图案转移

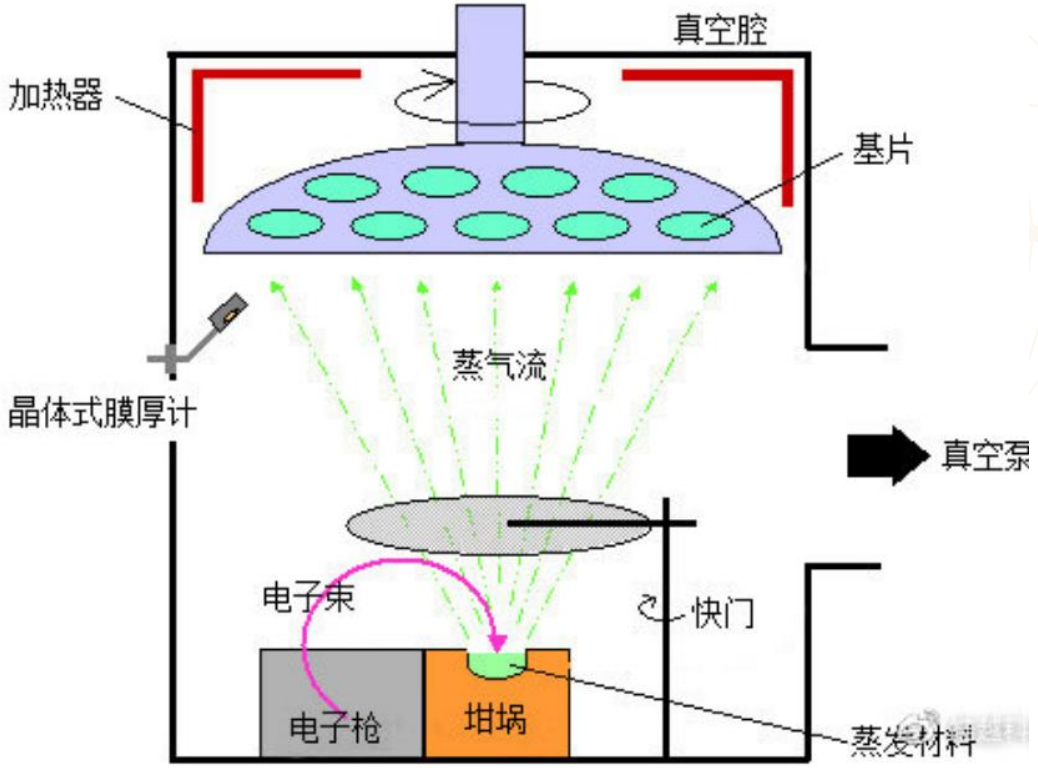


电子束光刻



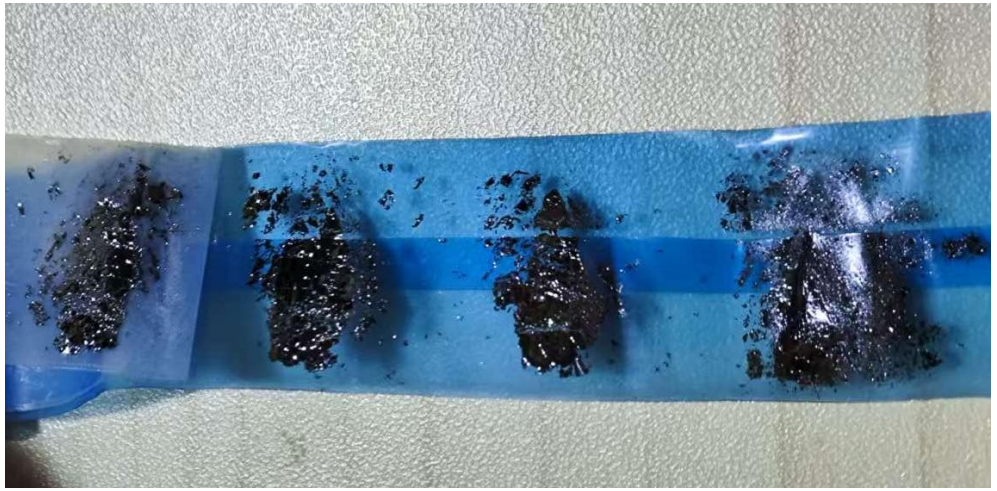
# 电极蒸镀



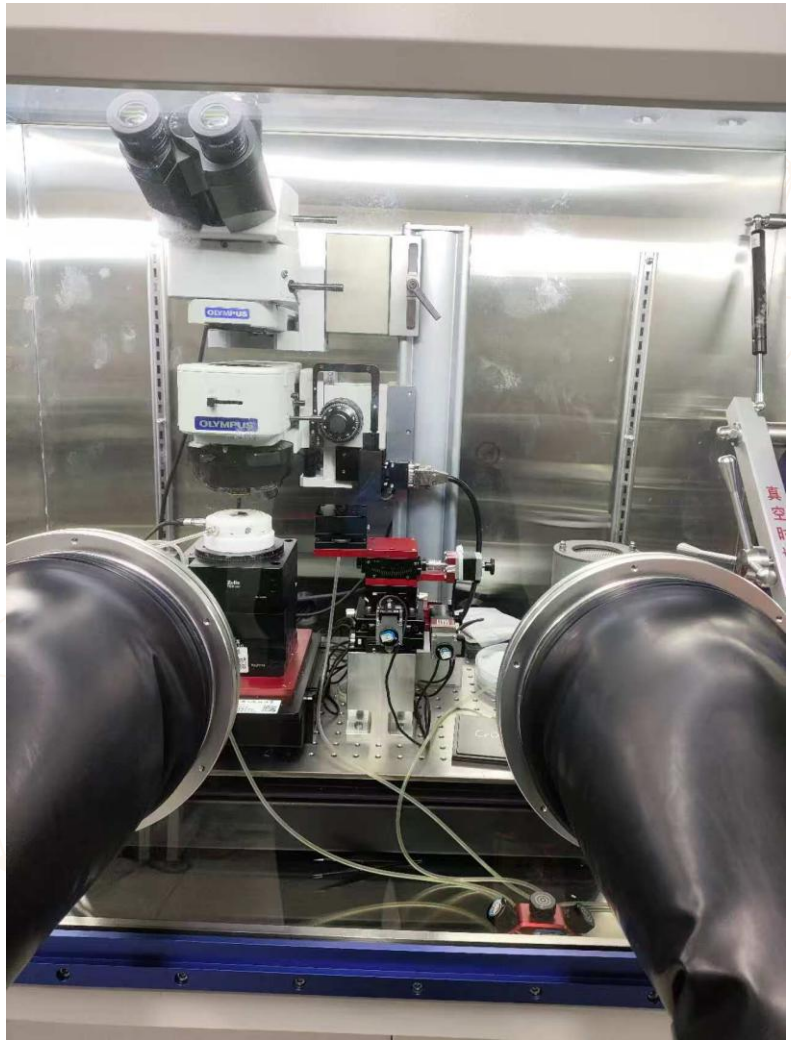



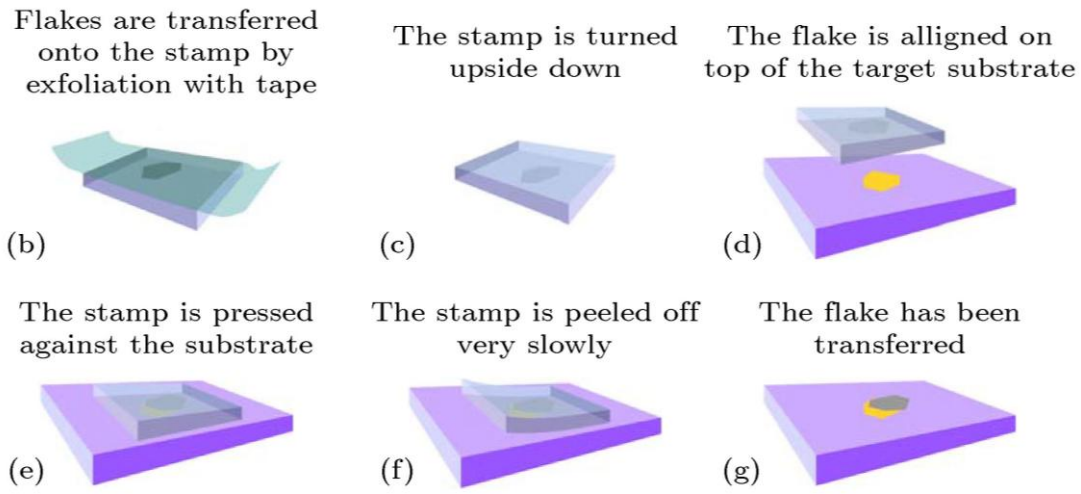

电子束蒸镀20nm Au 电极

# 材料剥离和转移



PDMS



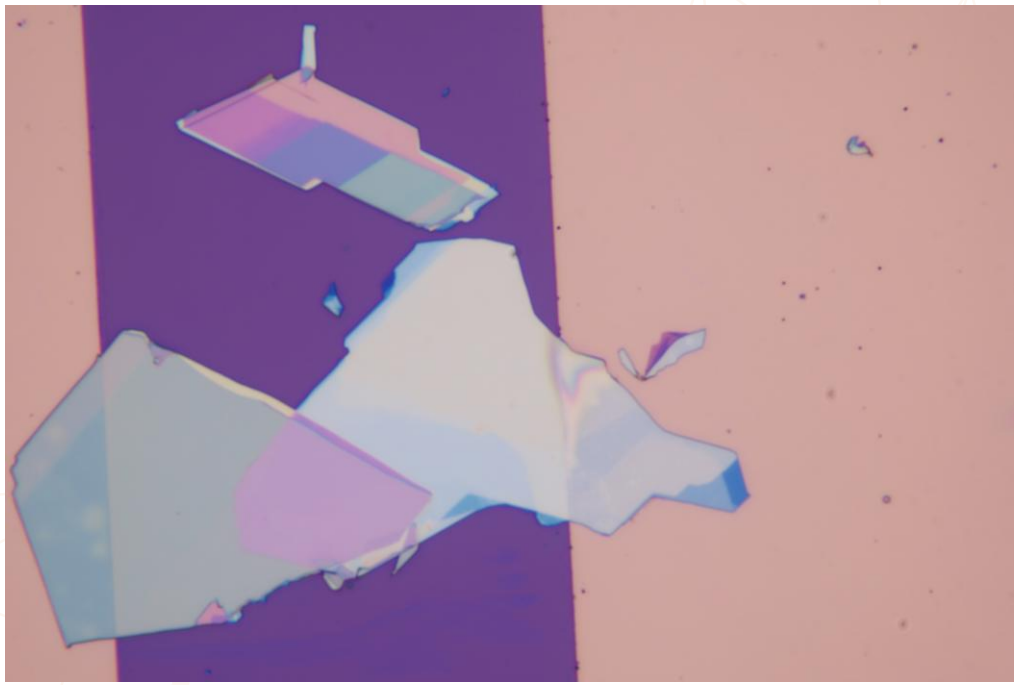


# 材料剥离和转移

SUSTech Southern University of Science and Technology

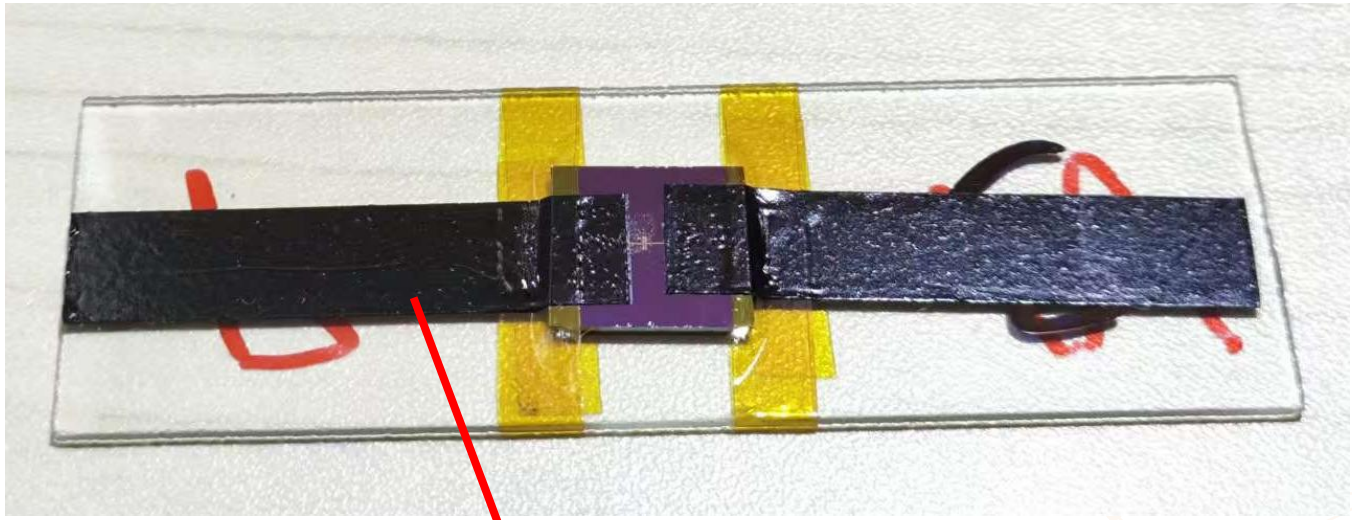


2D materials on PDMS

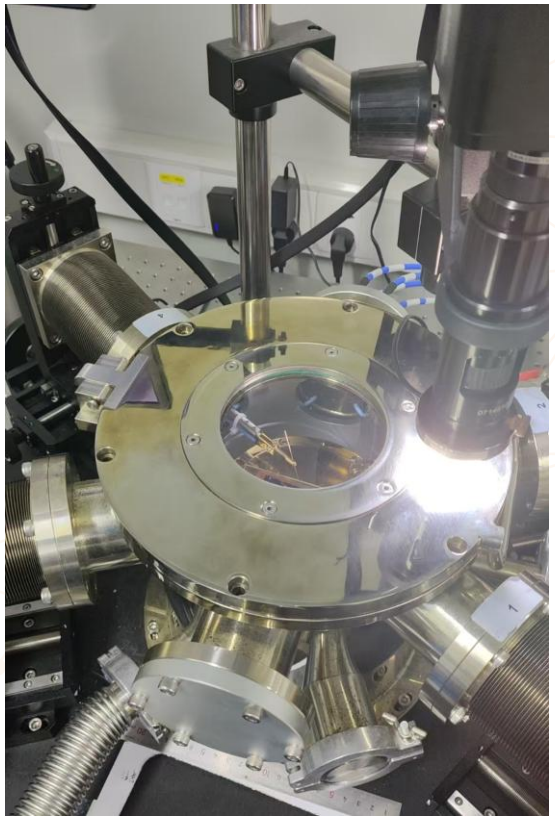


Materials transferred onto substrate

# 器件“封装”与测试



铝芯石墨导电胶带



探针台测试

SUSTech Southern University of Science and Technology

# 03

# 分析优化



# 不同材料组合的异质结

SUSTech Southern University of Science and Technology





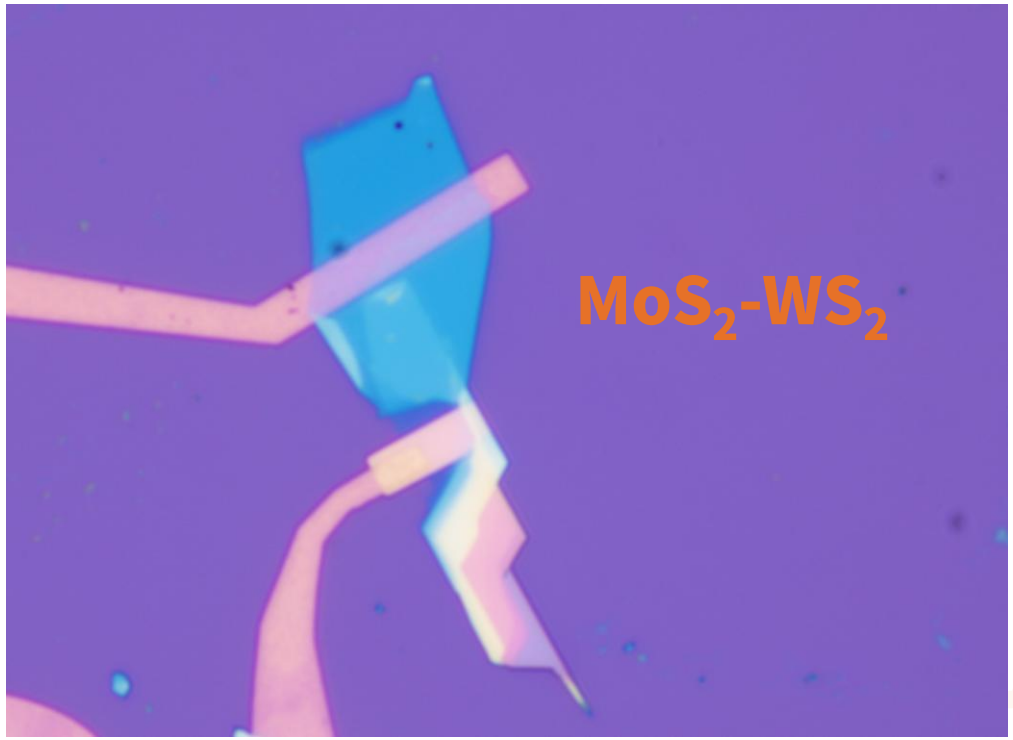


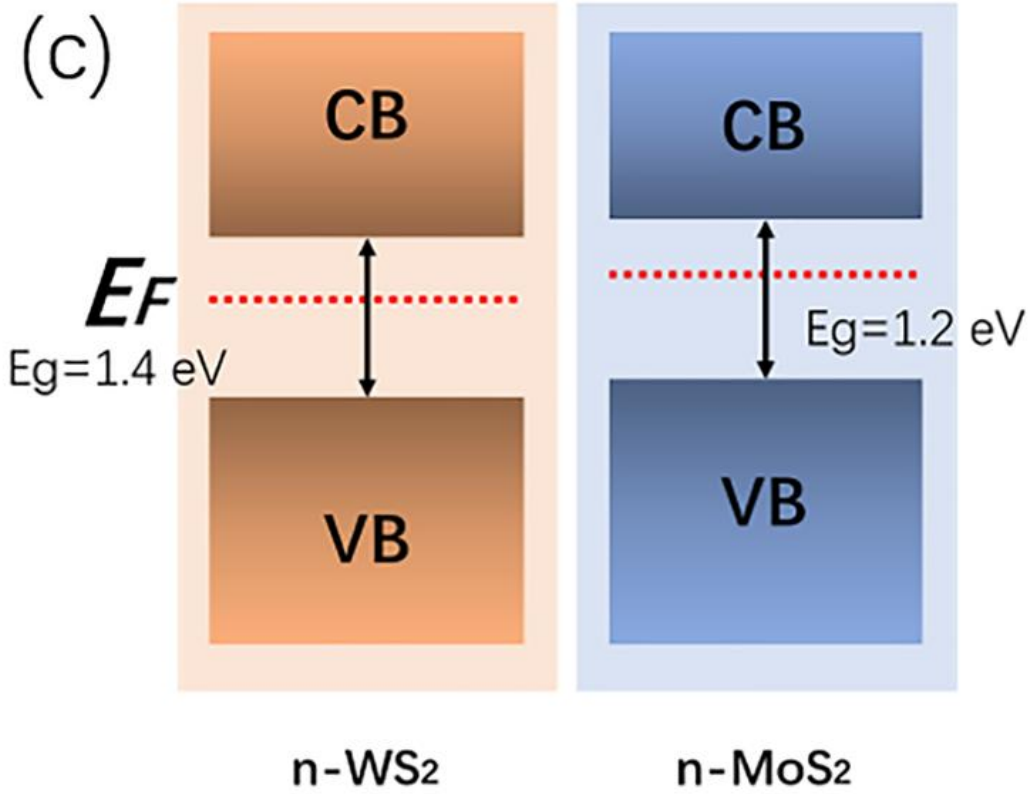





# $MoS_2$-$WS_2$ 异质结

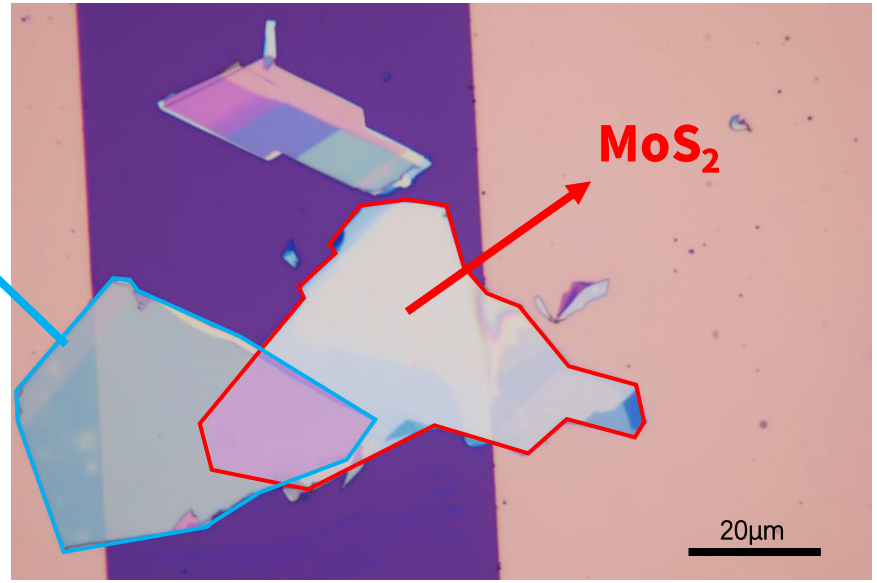
SUSTech Southern University of Science and Technology

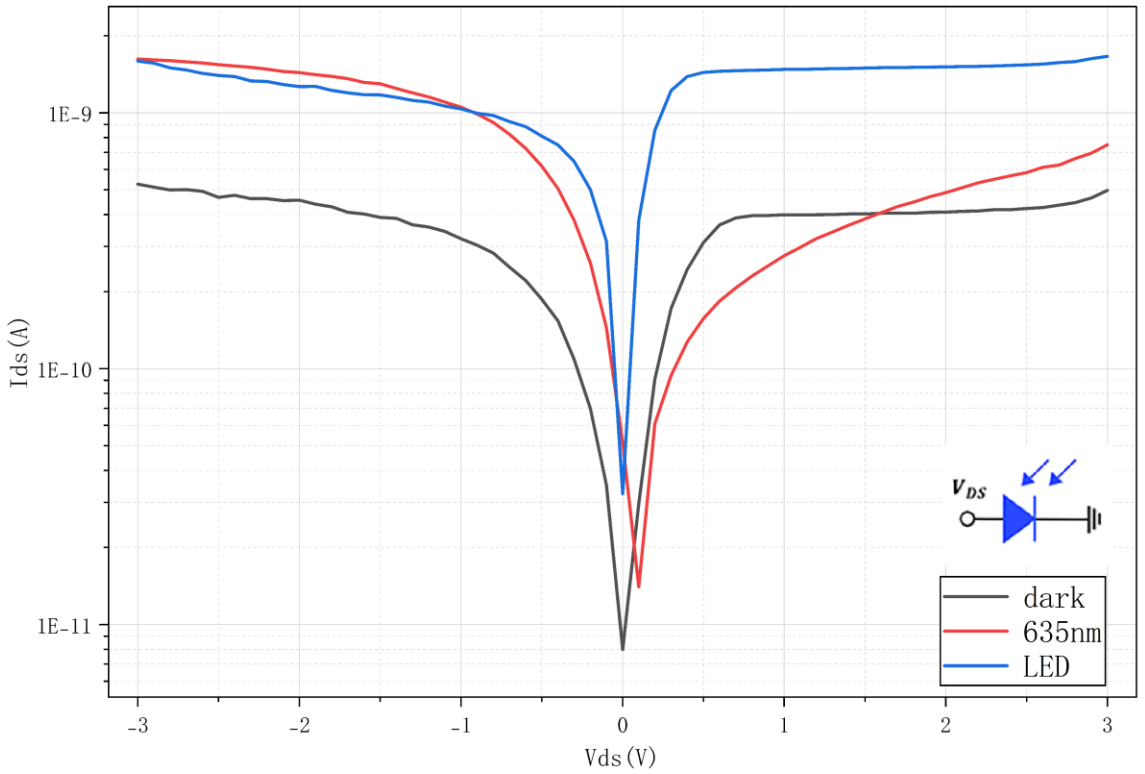



$MoS_2$-$WS_2$带隙较大，响应波长较短

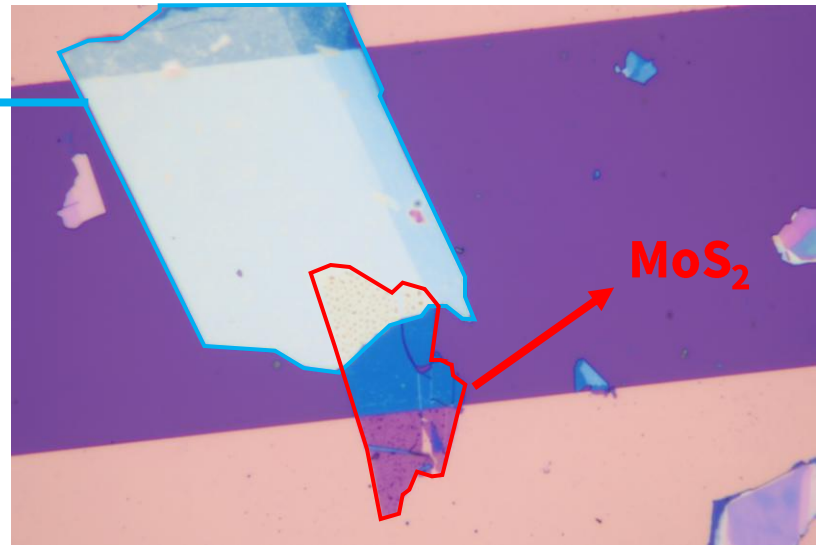
# 材料厚度选择

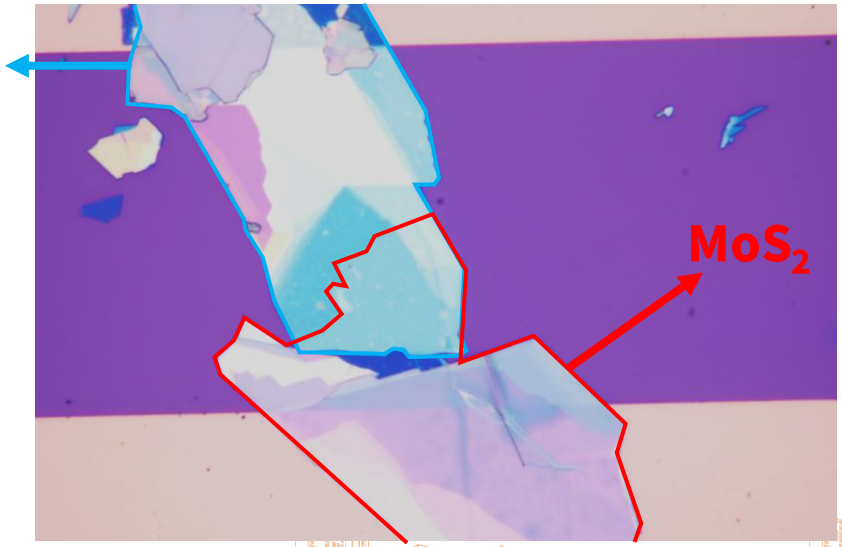


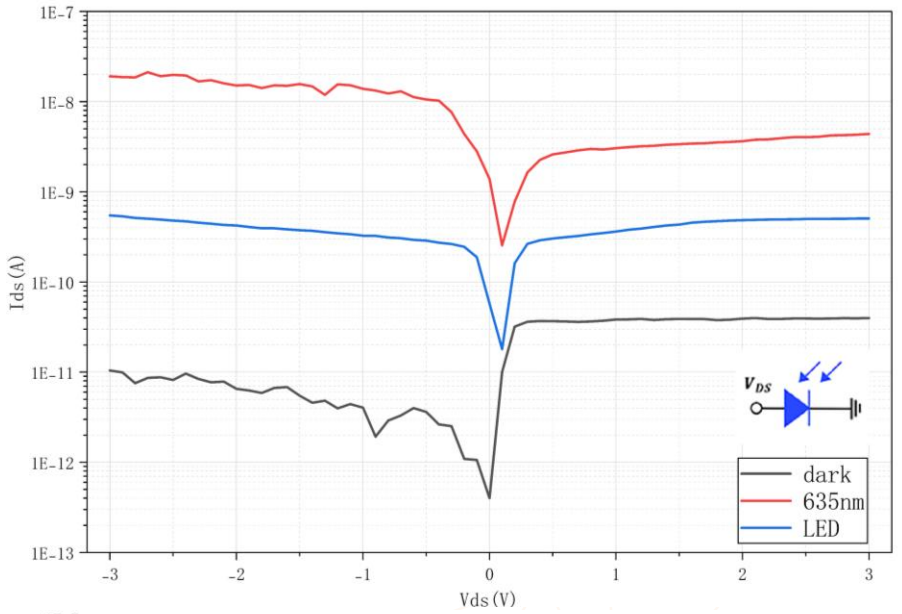



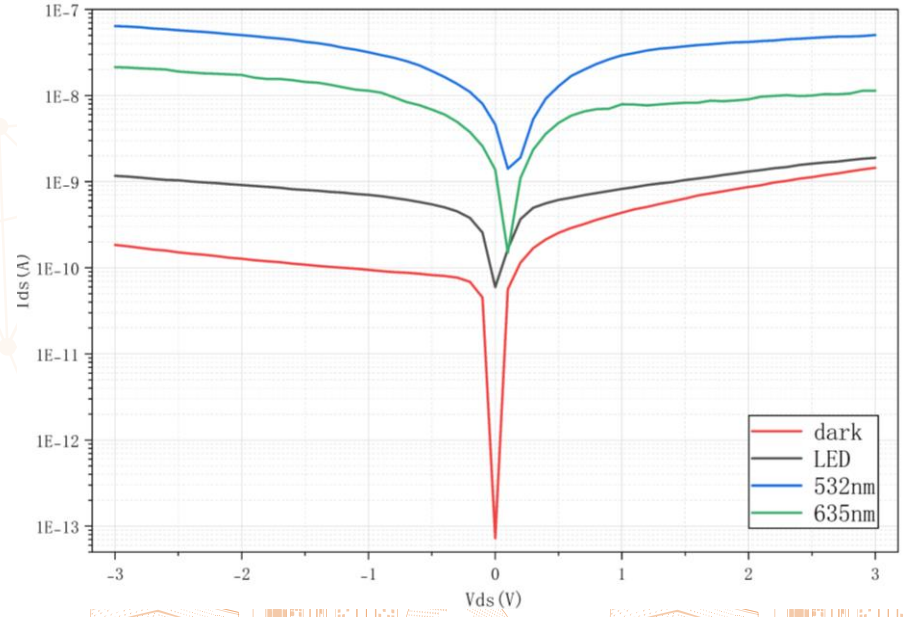

**材料厚度大，带隙较小，对可见光响应较弱，光电流较小**

# 材料厚度选择

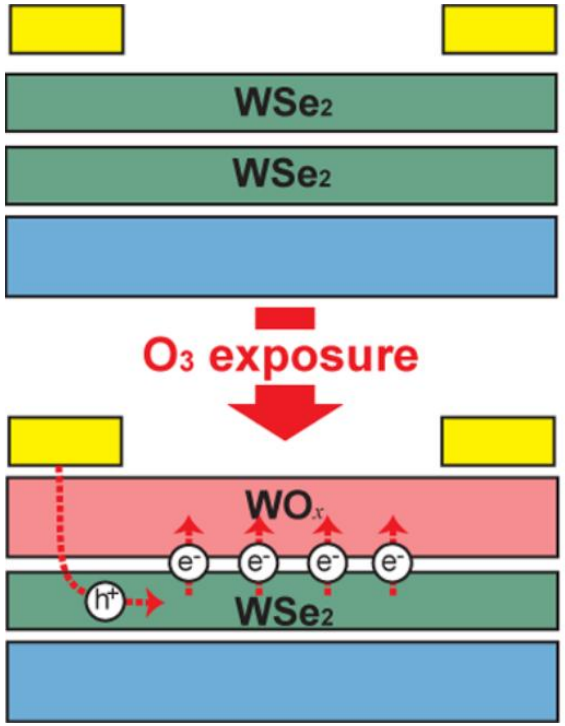

WS2
MoS2

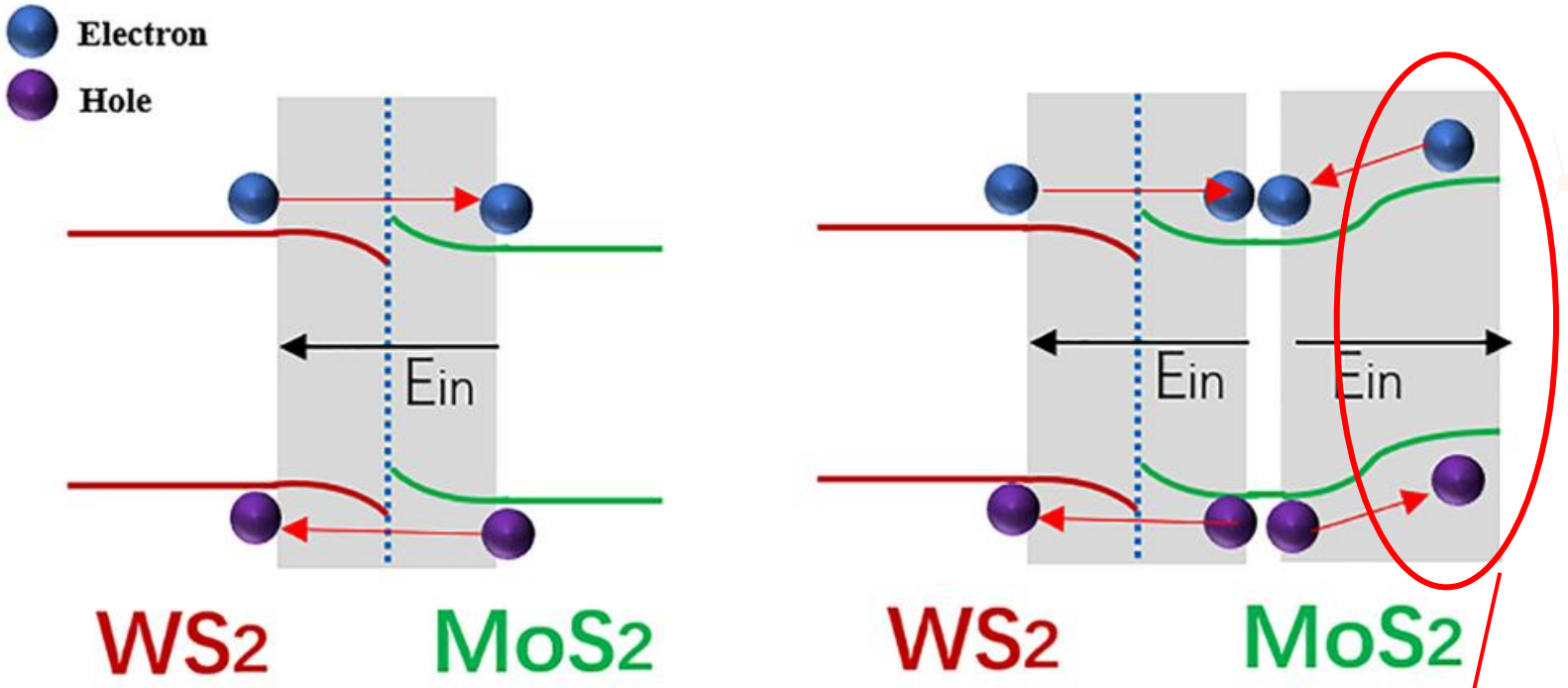

WS2
MoS2


1E-7
1E-8
1E-9
1E-10
1E-11
1E-12
1E-13
Ids (A)
-3
-2
-1
0
1
2
3
Vds (V)
$V_{DS}$
dark
635nm
LED


1E-7
1E-8
1E-9
1E-10
1E-11
1E-12
1E-13
Ids (A)
-3
-2
-1
0
1
2
3
Vds (V)
dark
LED
532nm
635nm

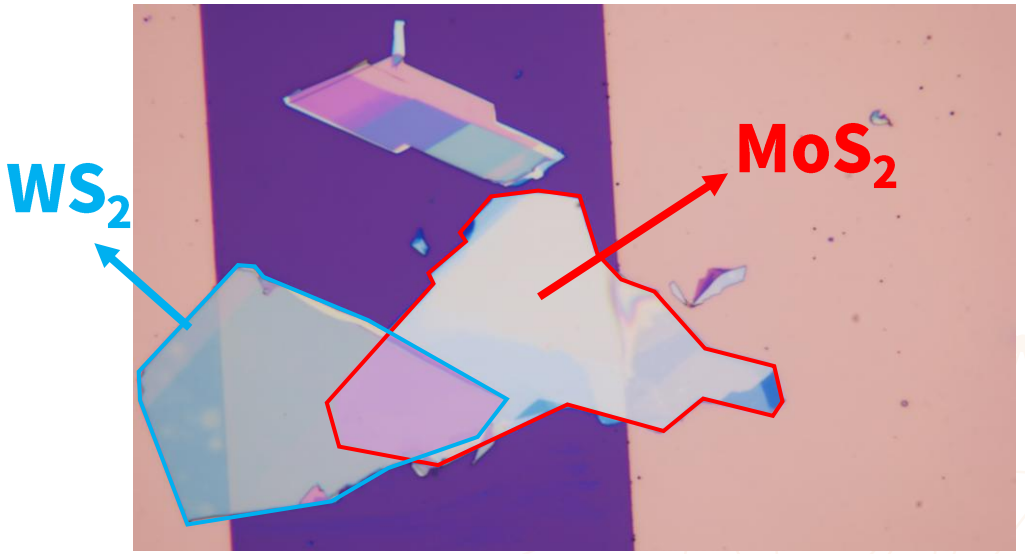
# P掺杂-臭氧处理

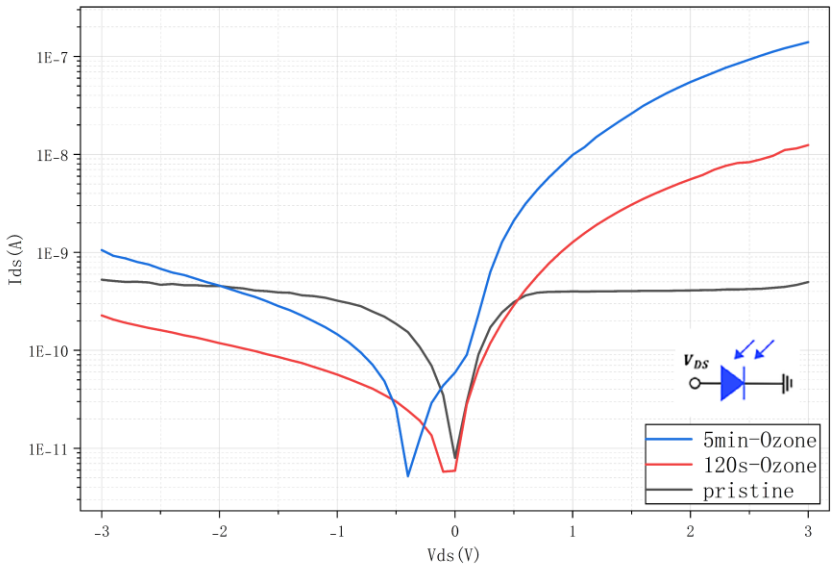


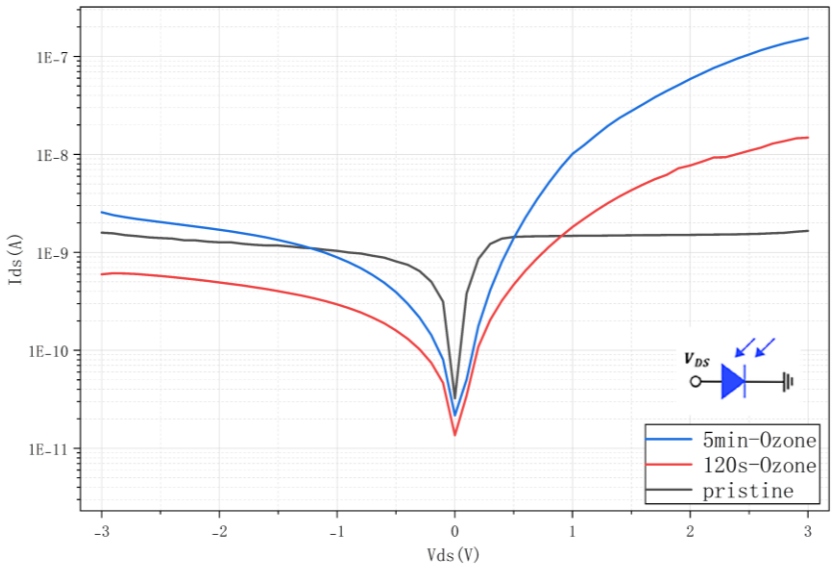

表层硫化钼被臭氧氧化成三氧化钼，抬高了下层氧化钼的费米能级，促进了载流子的分离

# P掺杂-臭氧处理

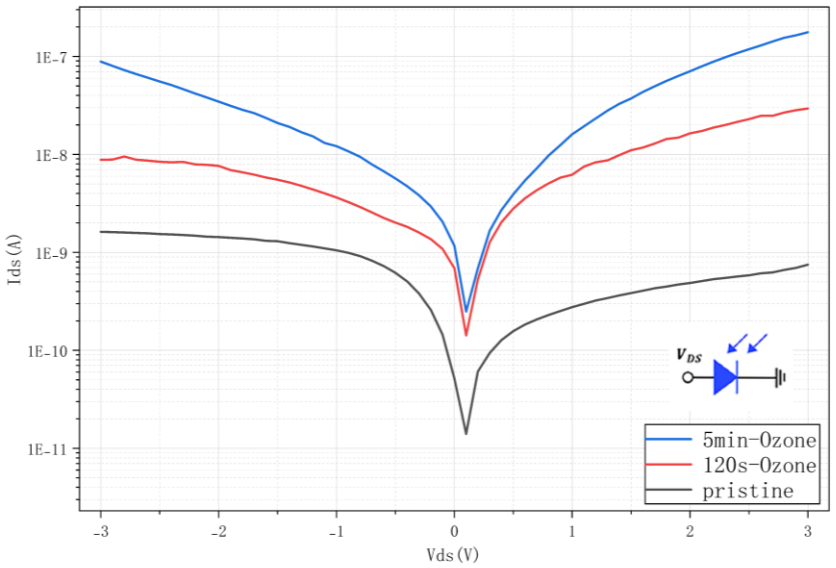



Dark




LED




632nm

# 优化迭代总结

SUSTech Southern University of Science and Technology

**电极：**

- 先转移材料再蒸镀电极➡先蒸镀材料再转移材料，避免了蒸镀电极时对材料的损伤，方便后续优化迭代，尝试不同的材料与组合

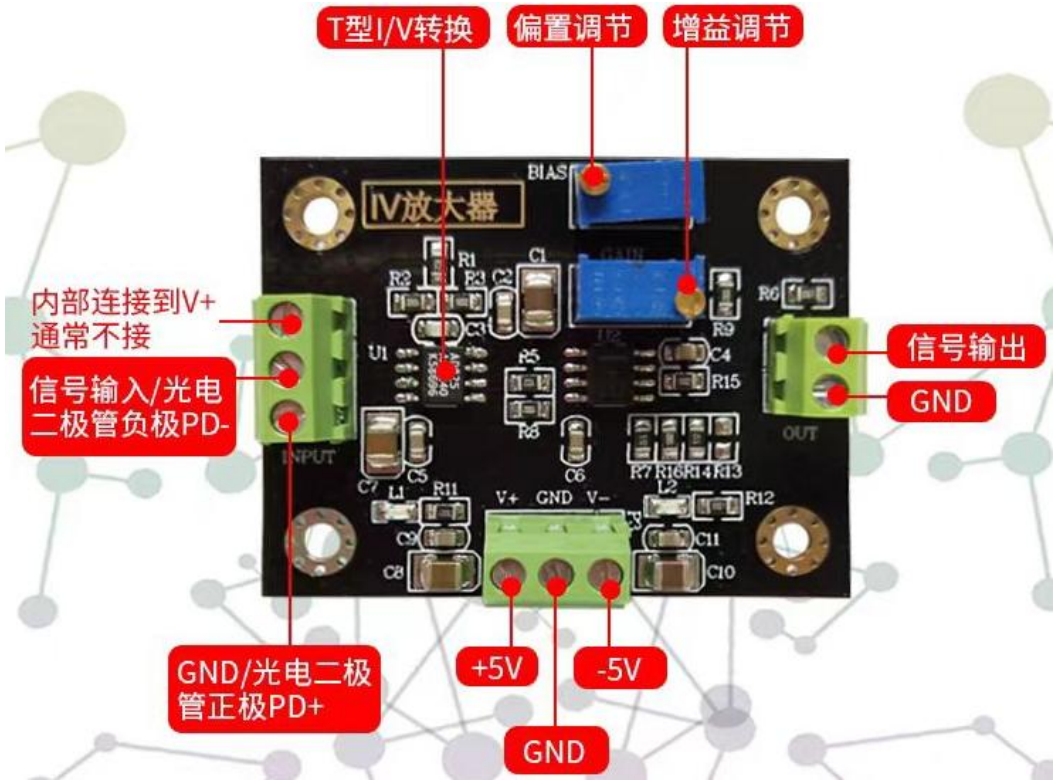
**厚度：**

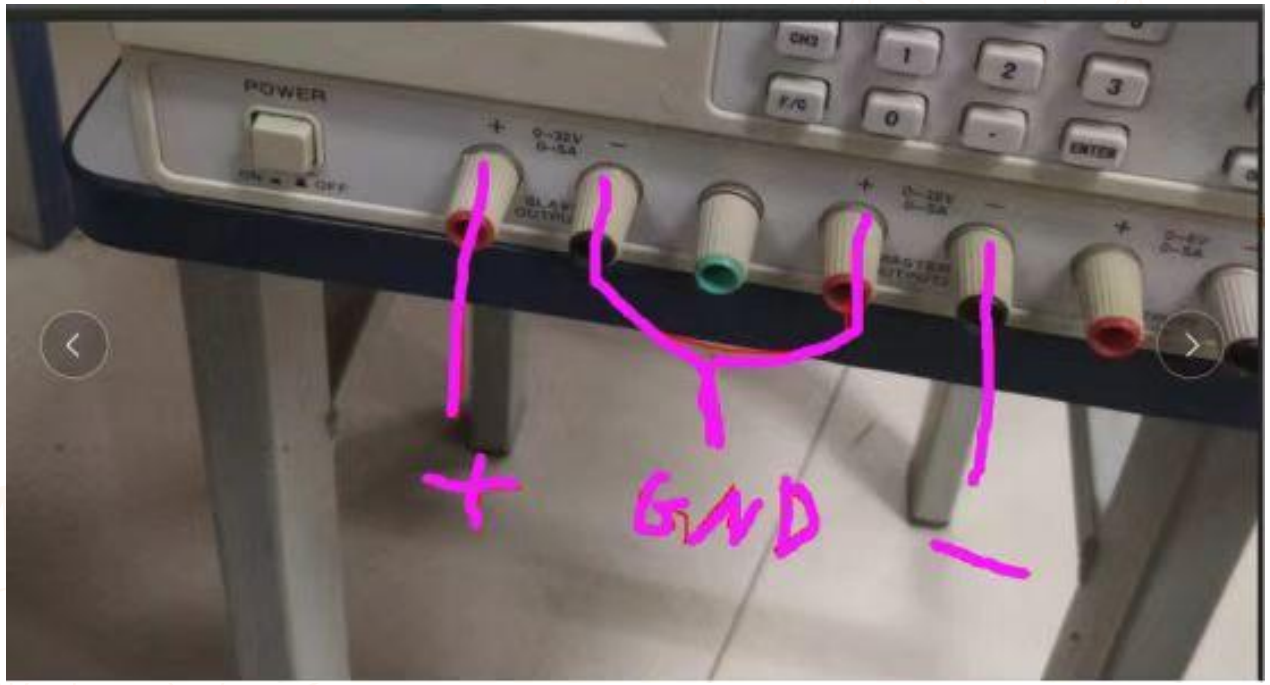
- 材料太厚➡带隙减小，对可见光不敏感，材料太薄➡光吸收率小，光电流小，难以探测

**臭氧P掺杂：**

- 实现高效P掺杂，促进了载流子的分离，增大了光电流

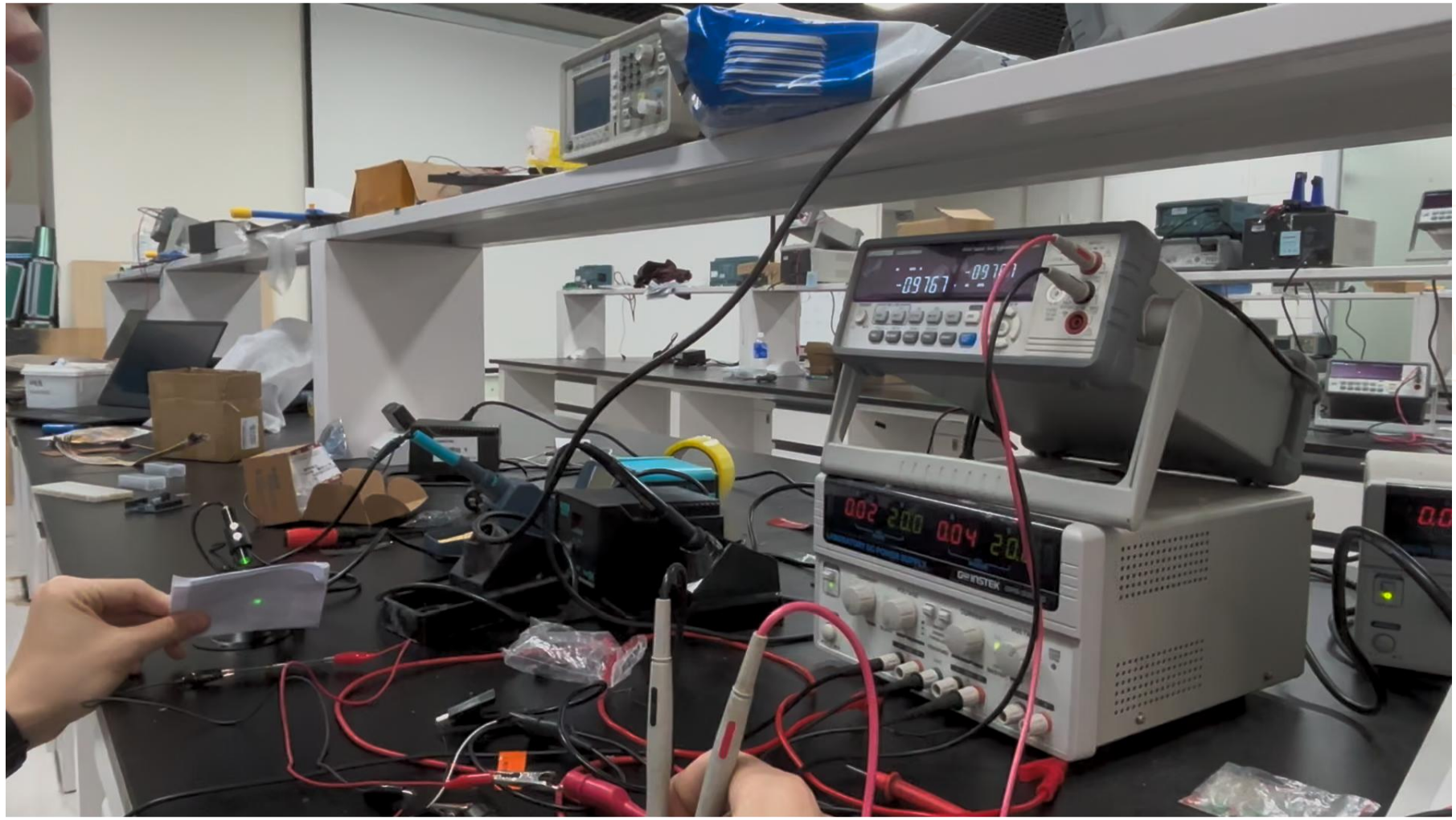
# 放大电路


T型I/V转换
偏置调节
增益调节
IV放大器
BIAS
内部连接到V+
通常不接
信号输入/光电
二极管负极PD-
信号输出
GND
GND/光电二极
管正极PD+
+5V
-5V
GND


POWER
+
GND
-

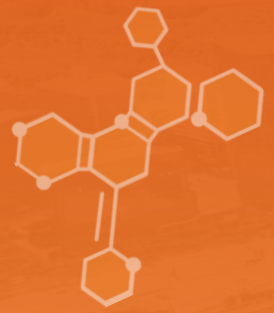
# 放大电路

SUSTech Southern University of Science and Technology





SUSTech Southern University of Science and Technology



# Thank you for your attention



# Reference

Liu, Y., Stradins, P., & Wei, S.-H. (2016). Van der Waals metal-semiconductor junction: Weak Fermi level pinning enables effective tuning of Schottky barrier. *Science Advances*, *2*(4), e1600069. https://doi.org/10.1126/sciadv.1600069

Yang, S., Wu, Z., Wang, S., Zheng, P., & Zhang, Y. (2024). Significantly Enhanced Photoresponse of Self-Powered 2D MoS2/WS2 Heterojunction Photodiode via F4-TCNQ Doping. *ACS Applied Electronic Materials*, *6*(5), 3374–3384. https://doi.org/10.1021/acsaelm.4c00192